

Private 5G latency: what transfers across locations?

Scenario and practical question

The TU Wien source contains 210 UDP trials: seven indoor locations, six configured rates and five repeats per setting. Each approximately 60-second iPerf run reports a mean one-way end-to-end delay. Three source-disclosed clock-wrap records are excluded without imputation, leaving 207 measured responses. The path includes campus routing, the 5G core/radio, customer equipment and a WiFi link.

We ask whether a compact equation reconstructs mean delay at a previously uncalibrated location using configured rate and achieved same-run throughput. This is an after-run diagnostic task. It is not a pre-run latency promise, radio-only model or causal congestion experiment. Native logs, units, exclusions and source anchors are audited in the retained research record.

Method and frozen equation

Five complete locations supply 147 development trials. Q4 and Q5 supply 60 reserved trials. Development uses leave-one-location-out validation; kernel tuning is nested within its training locations. Whole-location grouping keeps rates and repetitions together. Every candidate was frozen before the two confirmation scores were opened. The primary error is the unweighted mean of location-specific MAE in milliseconds.

Five substantive attempts test global traffic deficit, finite-setting response, within-setting correction, a high-load hinge and monotonicity. Attempt 003 was selected by development error and remains the reference despite its later failure to beat the flexible control.

For configured rate a and achieved rate b , both in Mbit/s, the equivalent centered equation is

$$\hat{L} = \max \{0, m_a + 1.179 (b - 1.05a)\} \quad \text{ms.}$$

Configured rate a (Mbit/s)	1	10	50	100	200	500
m_a (ms)	5.123	10.509	14.001	12.313	13.601	29.229

The coefficient 1.179 has units ms/(Mbit/s). The dimensionless 1.05 is only an algebraic recentering for readability; the executable state stores the uncentered exact coefficients. No interpolation between the six settings is admitted. All coefficients are estimated from development locations, with no held-out latency calibration.

Evidence and interpretation

Model	Development group MAE (ms)	Confirmation group MAE (ms)
Selected operating-point plus throughput rule	1.5094	2.1956
Operating-point medians without throughput	1.5841	1.9458
Monotonic operating-point medians	1.6181	1.9171
Nested flexible RBF control	1.6108	1.9165
Regularized source-family queue control	2.3271	3.0181
Achieved-throughput affine control	2.4882	2.9530

The selected rule improves the queue control by 27.3% on average, but is 14.6% worse than the flexible control. Its Q4/Q5 errors are 1.6631/2.7280 ms, compared with 1.7619/2.0710 ms for RBF. Thus the claimed gain from the within-setting slope fails overall confirmation. The later better controls remain comparators; they were not promoted after seeing confirmation.

The useful result is a bounded calibration recipe and an informative failure. Setting-specific response is more transferable here than a globally smooth queue approximation. However, adding a coefficient that looks like processing cost can improve development and degrade a new location. A 0.5 Mbit/s change in achieved-rate reporting changes the frozen equation by 0.5895 ms, making report precision relevant.

The difference $a - b$ is not measured packet loss: achieved rate is about 1.05-1.06 times configured rate in development. Neither that difference nor a fitted capacity pole identifies network physics. The source paper already studies queue-like delay and parameter identifiability. No new physical law or demonstrated operational benefit is claimed.

Use, limitations and evaluation

run.py and rules.json reproduce every frozen model. evidence/ contains predictions, aggregate metrics and both location results; TASK.md states the information budget and shared evaluator commands. Alternative equations are welcome under the same endpoint and cohort. Different endpoints require a separately reviewed task.

Only two reserved locations from one installation were tested. Clock synchronization is not independently verified; delay tails and independent deployment transfer remain unresolved. All current targets are now exposed, so future scoring is retrospective. The supported scope is an empirical reference with preserved counterexamples, not a universal communications law.

Sources

Bodur et al., Measurement-Driven Modeling of End-to-End Latency in an Indoor Private Standalone 5G Network, Network 6(3), 78 (2026), DOI 10.3390/network6030078. Pinned data: DOI 10.48436/hja53-s8y18,v1.0.0. Data: CC BY 4.0; source code: MIT. iPerf 2 user manual supplies report semantics. The complete targeted literature record is retained in PRIOR_ART.json.